The Network

The Multiplicative Graph of the Integers

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Abstract

The positive integers carry a natural network structure: nodes are integers, directed edges record divisibility, and prime factorization is the spanning forest. The BN flow is a random walk on this network, learning its topology one node at a time. Primes are the generators: every integer is a path through the prime nodes. Composites are intersections: they carry the correlation structure of the primes they share. RH is a statement about the spectral gap of this network: the arithmetic information at the primes propagates without loss to all frequencies of ζ. The network is spectrally complete if and only if RH holds.

1 The Integer Network

Definition 1 (The multiplicative network)**.** The *multiplicative network* $\mathcal{N}$ is the directed graph with:

- Vertex set: $\mathbb{Z}_{\geq 2}$.
- Edges: a directed edge $n \to m$ whenever $n \mid m$ and m/n is prime (i.e., m is obtained from n by multiplying by one prime).
- Root primes: vertices with in-degree 0.

Every vertex n has a unique factorization path from the root primes to n: $n = p_1^{a_1} \cdots p_k^{a_k}$, corresponding to the set of directed paths from $p_1, \ldots, p_k$ to n through the network.

Remark 2 (The network is not a tree)*.* The multiplicative network is not a tree because composites have multiple parents: $6 = 2 \cdot 3$ has parents $\{2, 3\}$ and $\{2, 3\}$, reached via $6/2 = 3$ and $6/3 = 2$. The composite 6 is an intersection of the paths from 2 and from 3. This intersection structure is exactly what the BN flow detects: when it adds $\rho_{1/6}$, it absorbs the correlation between the 2-path and the 3-path.

Definition 3 (The arithmetic Laplacian)**.** The *arithmetic Laplacian* of $\mathcal{N}$ is the operator $\mathcal{L}$ on $\ell^2(\mathbb{Z}_{\geq 2})$ defined by:

$$(\mathcal{L}f)(n) = \sum_{p \mid n} f(n/p) - \omega(n)\, f(n),$$

where $\omega(n)$ is the number of distinct prime divisors of n. $\mathcal{L}$ is the Laplacian of the undirected version of $\mathcal{N}$.

Proposition 4 (Eigenvectors are multiplicative functions)**.** *The multiplicative functions $n \mapsto n^{-s}$ are eigenvectors of $\mathcal{L}$ for every $s \in \mathbb{C}$:*

$$(\mathcal{L}[n^{-s}])(n) = \sum_{p|n} (n/p)^{-s} - \omega(n)\, n^{-s} = n^{-s} \Big(\sum_{p|n} p^s - \omega(n) \Big).$$

The eigenvalue is $\lambda_s(n) = \sum_{p|n} p^s - \omega(n)$, which depends on n (the operator is not translation-invariant). However, on prime nodes $n = p$: $\lambda_s(p) = p^s - 1$. On prime powers $n = p^k$: $\lambda_s(p^k) = p^s - 1$. The spectrum of $\mathcal{L}$ on primes is $\{p^s - 1 : p \text{ prime}\}$.

2 The BN Flow as Network Walk

Definition 5 (The network walk)**.** The BN flow at step N is a walk on $\mathcal{N}$: starting at node 2, adding one node at each step in the order $2, 3, 4, \ldots, N+1$. At each step, the walk absorbs the information carried by the new node and discards what was already known (Gram-Schmidt). The information at node n is $\rho_{1/n}$; the new information is $r_{n-1}^{\perp} \rho_{1/n}$, the component of $\rho_{1/n}$ orthogonal to all previous nodes.

Theorem 6 (Primes carry irreducible information)**.** *For prime p, the information $\rho_{1/p}$ is irreducible: it cannot be expressed as a combination of $\rho_{1/2}, \ldots, \rho_{1/(p-1)}$. This is equivalent to p being a prime node in $\mathcal{N}$ with no incoming edges from nodes $\leq p-1$ (since the edge $n \to p$ requires $n \mid p$ and p/n prime, which forces $n = 1$, outside the graph).*

Quantitatively ([1]): $\cos^2 \theta_{p-1} \geq c(\log p)^2/p$, measuring the irreducibility of the prime information.

Theorem 7 (Composites carry correlation)**.** *For composite $n = ab$ with $a, b > 1$, the information $\rho_{1/n}$ is largely redundant: $\cos^2 \theta_{n-1} \ll \cos^2 \theta_{p-1}$ for the nearest prime p. The composite information is the correlation $\mathrm{Corr}(\rho_{1/a}, \rho_{1/b})$ that was not captured by $\rho_{1/a}$ and $\rho_{1/b}$ independently. In the network: composite n lies at the intersection of the paths from its prime divisors, and its information measures how much the paths interact.*

3 The Spectral Gap

Definition 8 (Spectral completeness)**.** The network $\mathcal{N}$ is *spectrally complete* at frequency γ if the BN span $\mathcal{V}_\infty$ contains the function $x^{-1/2-i\gamma} 1_{(0,1)}$ in its L^2 closure. Spectral completeness at all frequencies means $\mathcal{V}_\infty$ is dense in $L^2(0,1)$, i.e., $d_\infty^2 = 0$.

Theorem 9 (Spectral completeness $\Leftrightarrow$ RH)**.** *The multiplicative network $\mathcal{N}$ is spectrally complete at all frequencies γ if and only if RH holds.*

- *Under RH: the zeros of ζ are all on $\mathrm{Re}(s) = 1/2$, so there is no off-critical zero to block any frequency ([2]). Every frequency γ is accessible from the network.*

- *Under ¬RH: the off-critical zero $\rho_0 = \sigma_0 + i\gamma_0$ blocks the frequency γ_0: the function $x^{-\rho_0}1_{(0,1)}$ is in $\mathcal{V}_\infty^\perp$ ([3]), meaning the network cannot propagate information at γ_0.*

Remark 10 (The spectral gap as an arithmetic defect). Spectral incompleteness at frequency γ_0 means: the multiplicative structure of the integers fails to propagate arithmetic information to the frequency γ_0 of ζ. This is a failure of the network's connectivity at γ_0.

In physical terms: the arithmetic Laplacian has a gap at frequency γ_0 — a mode of ζ that does not resonate with any multiplicative combination of the integers. RH eliminates all gaps: the network resonates with every mode of ζ.

4 The Network is the Proof

Theorem 11 (Network formulation of RH)**.** *RH is the statement:*

The multiplicative network of the integers is spectrally complete.

Equivalently:

1. *The arithmetic Laplacian $\mathcal{L}$ of $\mathcal{N}$ has no spectral gap at any frequency.*
2. *The random walk on $\mathcal{N}$ starting at the primes reaches every frequency in ζ's spectrum.*
3. *The BN flow, which traverses $\mathcal{N}$ node by node, eventually learns all of $1_{(0,1)}$.*
4. *The Euler product $\prod_p (1-p^{-s})^{-1}$ is spectrally faithful to $\zeta(s)$ on $\mathrm{Re}(s) = 1/2$.*

The network does not know about the zeros of ζ directly. It only knows primes and divisibility. RH says this is enough: the network's arithmetic is sufficient to capture the full analytic structure of ζ.

References

[1] Rincón, D. (2026). Prime. Preprint.

[2] Rincón, D. (2026). Möbius. Preprint.

[3] Rincón, D. (2026). The Density. Preprint.

[4] Rincón, D. (2026). Wisdom. Preprint.

[5] Rincón, D. (2026). The Explicit Formula. Preprint.